Subject: Mathematics | **PERIODIC TEST – 01 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 30**

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

A coin is tossed three times. The probability of getting heads and tails alternately (i.e. HTH or THT) is:

(a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) $\frac{3}{8}$ (d) $\frac{1}{2}$

Working & Calculation:

- Tossing a coin 3 times yields $n(S) = 2^3 = 8$ equally likely outcomes:

$$S = \{\text{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}\}$$

- The alternating outcomes are $E = \{\text{HTH, THT}\} \implies n(E) = 2$:

$$P(\text{alternating heads and tails}) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{4}$

Question 2 [1 Mark]

A die is thrown once. The probability of getting an even prime number is:

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) 0

Working & Calculation:

- A standard die has $S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$.
- Prime numbers on the die are $\{2, 3, 5\}$. The only even prime number is 2:

$$E = \{2\} \implies n(E) = 1 \implies P(\text{even prime}) = \frac{1}{6}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{6}$

Question 3 [1 Mark]

From a pack of 52 cards, a card is drawn at random. The probability that it is neither an ace nor a king is:

(a) $\frac{11}{13}$ (b) $\frac{12}{13}$

(c) $\frac{9}{13}$ (d) $\frac{10}{13}$

Working & Calculation:

- Total cards in a standard deck: $n(S) = 52$.
- There are 4 aces and 4 kings $\implies$ Total aces and kings $= 4 + 4 = 8$ cards.
- Favourable cards (neither an ace nor a king) $= 52 - 8 = 44$:

$$P(\text{neither ace nor king}) = \frac{44}{52} = \frac{11}{13}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{11}{13}$

Question 4 [1 Mark]

A bag contains 4 red, 3 blue, and 2 yellow balls. One ball is drawn at random from the bag. The probability of getting a yellow ball is:

(a) $\frac{2}{9}$ (b) $\frac{1}{3}$

(c) $\frac{4}{9}$ (d) $\frac{5}{9}$

Working & Calculation:

- Total number of balls: $n(S) = 4 + 3 + 2 = 9$.
- Number of yellow balls: $n(\text{yellow}) = 2 \implies P(\text{yellow ball}) = \frac{2}{9}$.

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{2}{9}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): The probability of getting a number greater than 6 in a single throw of a standard die is 0.
Reason (R): Getting a number greater than 6 on a standard die is a sure event.

Options:

(a) **Both (A) and (R) are true, and (R) is the correct explanation of (A).**
(b) **Both (A) and (R) are true, but (R) is not the correct explanation of (A).**
(c) **(A) is true, but (R) is false.**
(d) **(A) is false, but (R) is true.**

Step-by-Step Logical Analysis:

1. **Assertion (A):** A standard die only contains faces $\{1, 2, 3, 4, 5, 6\}$. There are no faces > 6. This is an impossible event with probability 0. $\implies$ **(A) is TRUE.**
2. **Reason (R):** An event that cannot occur is termed an **impossible event** (probability 0), whereas a **sure event** has probability 1. Therefore, claiming this is a sure event is mathematically incorrect. $\implies$ **(R) is FALSE.**

CORRECT ANSWER: OPTION (c) $\longrightarrow$ (A) is true, but (R) is false.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

Question 6 [2 Marks]

A coin is tossed three times. List the sample space and find the probability of getting at least two heads.

Sample Space: $S = \{\text{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}\} \implies n(S) = 8.$

At least two heads (2 or 3 heads): $E = \{\text{HHT, HTH, THH, HHH}\} \implies n(E) = 4$:

$$P(\text{at least two heads}) = \frac{n(E)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Answer: $\mathbf{P(\text{at least two heads}) = \frac{1}{2}}$

Question 7 [2 Marks]

A box contains cards numbered from 11 to 60. A card is drawn at random from the box. Find the probability that the number on the card is: (i) divisible by 5 (ii) a perfect square.

Card Count: Total cards numbered 11 to 60: $n(S) = 60 - 11 + 1 = 50$ cards.

(i) Divisible by 5 [1 Mark]

Multiples of 5 are $\{15, 20, 25, 30, 35, 40, 45, 50, 55, 60\} \implies n(E_1) = 10 \implies P = \frac{10}{50} = \frac{1}{5}.$

Answer (i): $\mathbf{P(\text{divisible by 5}) = \frac{1}{5}}$

(ii) A perfect square [1 Mark]

Perfect squares are $\{16, 25, 36, 49\} \implies n(E_2) = 4 \implies P = \frac{4}{50} = \frac{2}{25}.$

Answer (ii): $\mathbf{P(\text{perfect square}) = \frac{2}{25}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

Question 8 [3 Marks]

12 defective pens are accidentally mixed with 132 good ones. It is not possible to just look at a pen and tell whether or not it is defective. One pen is taken out at random from this lot.
(i) Determine the probability that the pen taken out is a good one.
(ii) What is the probability that the pen taken out is defective?
(iii) If 6 good pens are added to this lot, what will be the new probability of drawing a defective pen?

Initial Setup: Total pens $n(S) = 12 + 132 = 144.$

(i) Probability of drawing a good pen [1 Mark]

Number of good pens $= 132 \implies P(\text{good pen}) = \frac{132}{144} = \frac{11}{12}.$

Answer (i): $\mathbf{P(\text{good pen}) = \frac{11}{12}}$

(ii) Probability of drawing a defective pen [1 Mark]

Number of defective pens $= 12 \implies P(\text{defective pen}) = \frac{12}{144} = \frac{1}{12}.$

Answer (ii): $\mathbf{P(\text{defective pen}) = \frac{1}{12}}$

(iii) New probability of drawing a defective pen after adding 6 good pens [1 Mark]

New total pens $= 144 + 6 = 150$. Defective pens remain 12 $\implies P'(\text{defective}) = \frac{12}{150} = \frac{2}{25} = 0.08.$

Answer (iii): $\mathbf{P'(\text{defective pen}) = \frac{2}{25} = 0.08}$

Question 9 [3 Marks]

A square dartboard has sides of length 20 cm. Inside this board, a circle of diameter 14 cm is drawn. A dart is thrown at random and hits the dartboard. Taking $\pi = \frac{22}{7}$, calculate the probability that the dart lands: (i) inside the circle (ii) outside the circle (iii) Find the ratio of the probability of landing inside the circle to landing outside.

Geometric Areas: Side of square $s = 20$ cm $\implies$ Area of square $A_S = 20^2 = 400$ cm^2.
Circle diameter $d = 14$ cm $\implies r = 7$ cm $\implies$ Area of circle $A_C = \pi r^2 = \frac{22}{7} \times 7^2 = 154$ cm^2.

(i) Probability of landing inside the circle [1 Mark]

$$P(\text{inside circle}) = \frac{A_C}{A_S} = \frac{154}{400} = \frac{77}{200} = 0.385.$$

Answer (i): $\mathbf{P(\text{inside circle}) = \frac{77}{200} = 0.385}$

(ii) Probability of landing outside the circle [1 Mark]

$$\text{Area outside} = 400 - 154 = 246 \text{ cm}^2 \implies P(\text{outside}) = \frac{246}{400} = \frac{123}{200} = 0.615.$$

Answer (ii): $\mathbf{P(\text{outside circle}) = \frac{123}{200} = 0.615}$

(iii) Ratio of probability of landing inside to landing outside [1 Mark]

$$\text{Ratio} = \frac{P(\text{inside})}{P(\text{outside})} = \frac{154/400}{246/400} = \frac{154}{246} = \frac{77}{123}.$$

Answer (iii): Ratio = 77 : 123

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

Question 10 [5 Marks]

Two dice, one blue and one grey, are thrown at the same time.
(i) Write down the total number of elementary outcomes in the sample space.
(ii) Complete the probability distribution for the sum of the two numbers turning up:
(iii) A student argues that there are 11 possible sums (2 to 12), and therefore each has probability $\frac{1}{11}$. Do you agree with this argument? Justify your answer.
(iv) What is the probability that the sum of the numbers is at least 10?

(i) Total elementary outcomes [1 Mark]

Each die has 6 faces $\implies n(S) = 6 \times 6 = 36$ elementary outcomes.

Answer (i): n(S) = 36 outcomes

(ii) Completed Probability Distribution Table [1.5 Marks]

Sum on 2 dice	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(iii) Justification of student's argument [1.5 Marks]

No, we do not agree with the argument. The 11 possible sums are **not equally likely**. For instance, sum 7 has 6 favourable outcomes ($P = 6/36$), whereas sum 2 has only 1 favourable outcome ($P = 1/36$). The formula $1/11$ applies only to equally likely outcomes, which does not hold here.

Answer (iii): Disagree. The 11 outcomes are not equally likely.

(iv) Probability that sum is at least 10 [1 Mark]

Sums ≥ 10 are $\{10, 11, 12\}$. Favourable outcomes: $3 + 2 + 1 = 6 \implies P(\text{sum} \geq 10) = \frac{6}{36} = \frac{1}{6}$.

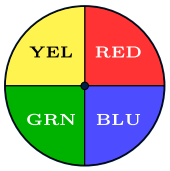
Answer (iv): $\mathbf{P(\text{sum} \geq 10) = \frac{6}{36} = \frac{1}{6}}$

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

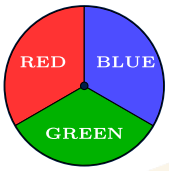
Question 11 (Case Study Analysis) [4 Marks]

A youth club organizes a color mixing stall at their annual carnival. The game utilizes two independent circular spinners:
· Spinner 1 has 4 equal sectors: Red (R), Yellow (Y), Green (G), Blue (B).
· Spinner 2 has 3 equal sectors: Red (R), Green (G), Blue (B).
A player spins both spinners. By color theory, mixing Red and Blue creates Purple. A player wins if one spinner lands on Red and the other spinner lands on Blue. Based on the scenario, answer the following:

Spinner 1 (4 Sectors)



Spinner 2 (3 Sectors)



Combinatorial Outcomes: Total combined outcomes $n(S) = 4 \times 3 = 12$.

(i) Total number of combined outcomes when both spinners are spun [1 Mark]

Each outcome is an ordered pair (Spinner 1, Spinner 2) $\implies n(S) = 4 \times 3 = 12$.

Answer (i): $\mathbf{n(S) = 4 \times 3 = 12}$ combined outcomes

(ii) List winning pairs and find probability player wins (makes purple) [1 Mark]

Winning requires Red and Blue $\implies E = \{(\text{Red, Blue}), (\text{Blue, Red})\} \implies n(E) = 2$:

$$P(\text{makes purple}) = \frac{n(E)}{n(S)} = \frac{2}{12} = \frac{1}{6}$$

Answer (ii): Winning Pairs: $\{(\text{R, B}), (\text{B, R})\}$, $\mathbf{P(\text{wins}) = \frac{1}{6}}$

(iii) Probability that both spinners show the same color [1 Mark]

Common colors are Red, Green, and Blue $\implies E_{\text{same}} = \{(\text{R, R}), (\text{G, G}), (\text{B, B})\} \implies n(E_{\text{same}}) = 3$:

$$P(\text{same color}) = \frac{3}{12} = \frac{1}{4}$$

Answer (iii): $\mathbf{P(\text{same color}) = \frac{1}{4}}$

(iv) Probability that neither spinner lands on Green [1 Mark]

Non-green on Spinner 1 has 3 choices; non-green on Spinner 2 has 2 choices $\implies 3 \times 2 = 6$ pairs:

$$P(\text{neither Green}) = \frac{6}{12} = \frac{1}{2}$$

Answer (iv): $\mathbf{P(\text{neither lands on Green}) = \frac{1}{2}}$

— END OF WORKED SOLUTIONS (TEST–07) · MATHS MANTHAN TUITION CLASSES —